



<Code Club JUSL/>



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Sprinklr Product Engineer Intern

Internship Acceptance Guide

A complete guide for aspiring interns

Student's Experience & Suggestions

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Anay Saha

Information Technology

Name	Anay Saha
Department	Information Technology
Year of Grad.	2027

Experience

The OA consisted of 3 questions, and we had to solve any 2 to move to the next round.

In my 1st technical round, the interviewer first asked me to introduce myself. I gave a short introduction — my name, college, and that's it.

1st Question:

He asked me to create a MedianFinder class with two methods:

addNum() – adds a number into a data structure.

findMedian() – returns the median.

(This is the same as the LeetCode problem "Find Median from Data Stream.")

2nd Question (Puzzle):

We have n flasks, where n-1 flasks contain water and 1 flask contains acid. We have an infinite number of litmus papers. Find the minimum number of litmus papers required to identify the acidic flask.

I couldn't solve it, but I explained my thought process. Even though I wasn't getting the answer, I asked for hints and tried working through them, but still couldn't solve it.

In the 2nd technical round, the first question was:

he asked about 2-3 dbms questions and then this was followed

1st Question:

Create an Autocomplete class that stores a set of names. When a user types a string s , it should return suggestions of names starting with s , showing the top 10 most frequent matches.

This was based on a Trie. I struggled with one edge case and asked the interviewer for an example where my approach would fail, but he didn't give one.

2nd Question:

We are given n inequality equations in the form $[v1, v2, type]$, where:

$type = 0$ means $v1 == v2$

$type = 1$ means $v1 != v2$

We need to return true if we can assign integers to the variables to satisfy all equations.

Example:

$[a, b, 0]$

$[a, b, 1] \rightarrow \text{false}$

$[a, b, 0]$

$[c, a, 0] \rightarrow \text{true} (a = b = c = 1)$

$[a, b, 0]$

$[c, b, 1] \rightarrow \text{true} (a = b = 1, c = 2)$

$[a, b, 0]$

$[a, b, 1] \rightarrow \text{false} (\text{contradiction})$

My first approach was using a graph:

For all $type = 0$ pairs, connect the nodes (same group).

Then for each $type = 1$ pair, check if both variables are connected (if yes $\rightarrow$ contradiction).

This approach was $O(n^2)$, so the interviewer asked me to optimize it. I then used Disjoint Set Union (DSU) to solve it more efficiently.

He said it was fine, and the interview ended.

In the HR round, it was more of a casual conversation — talking about myself and what I know about the company. Many people were asked about their projects in the technical rounds, but in my case, I was only asked about my tech stack during the HR round.

Prep. Tips

If you don't know take some time to think try harder ask for some hints but not always one round one hint at max explain yourself like you are explaining to a newbie be clear and understandable and have a positive attitude don't panic that's all.

Additional Notes

Try to explain out loud to yourself even when you are solving alone this will make it a habit.

✓ **Selected**

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Experience

OA Round : The OA consisted of 3 DSA questions and was conducted on HackerEarth through a proctored Smartbrowser application. The first question was a Sliding Window question exactly similar to this LC question Maximum Points You Can Obtain from Cards - LeetCode . The 2nd question was a DP on Trees where each edge had a positive or negative weight and we have to maximize the sum of selected weights taken but we cannot pick adjacent edges. The 3rd question was a PnC type question where we have to count how many numbers having either a or b as digits and the number formed by summing the digits also has only either a or b. I was able to solve the first two entirely but could not reach the optimal solution for the 3rd.

Technical Interview 1 : The interview was conducted on Microsoft Teams and I was asked to code in CodeShare, an online collaborative code editor. The interviewer introduced himself and asked for my short introduction, then he proceeded to ask about my projects mentioned in my resume. I had used MongoDB so questions were related to indexing in MongoDB, need for databases over file systems, what problem does my application solve, is auth taken care of while logging in, how is the Map api used(I had used an open-source map), why did I use NextJs instead of plain React. He asked whether I knew OOPS and proceeded to ask me questions on Polymorphism, its types, which one is needed where. There was one DSA question where a startString and an endString was given and a String bank and we have to find the minimum number of steps required to convert from startString to endString by changing 1 character each time. I solved the question using BFS and explained why a queue is needed in it. He asked about time complexity and seemed pleased with the approach. The interviewer was very supportive.

Technical Interview 2 : This round was all about two DSA questions. The first one was we were given a binary matrix (mxn) with each row sorted and asked to find the row number with max 1's . I gave a binary search on each row approach with $O(m \log n)$ complexity but the interviewer hinted me towards a $O(n+m)$ complexity approach which I was able to code up using a pointer on the first 1 in each column greedy solution. Then the interviewer proceeded to ask me on Binary trees and Binary Search Trees and their traversals and how to construct a unique tree from them. I was given inorder and preorder traversals and asked to construct the postorder from them. I coded a recursive approach which had some flaws in some edge cases but the interviewer seemed overall pleased with the approach. However I could not see the interviewer in this round because he had not turned on his video.

HR Interview : The interviewer asked about my day and how the other two rounds went. He asked me for a short introduction and what I knew about Sprinklr. He asked whether I had a conversation with any employee before the interviews and then proceeded to ask about my projects and academics a bit. He asked what I would do in the 9 months before the internship started and I told him that I would improve my skills. He asked about my weaknesses and how I am trying to improve. The interviewer was friendly and seemed pleased with the answers.

Prep. Tips

I recommend being thorough with Striver's A2Z DSA sheet and solving it as many times as possible. The DSA questions were all some variation of this sheet's questions. Also you should know properly about your own projects and why you used a particular tech stack. I recommend Algozenith's placement resources for the core CS topics. They have free notes for DBMS, CN , OS and OOPS.

✓ **Selected**